

# Phishing inbox

Revise 360 lesson plan — 60 minutes

|                 |                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Unit            | OCR J277 1.4 Network security — Paper 1, Computer systems                                                         |
| Lesson          | Phishing inbox (lesson 4 of the unit)                                                                             |
| Lesson ID       | ns-104                                                                                                            |
| Experience ID   | ns-104                                                                                                            |
| Plan ID         | PLAN-NS-L04                                                                                                       |
| Content version | 0641c48a7420                                                                                                      |
| Website         | <a href="https://revise360.co.uk/experience.html?id=ns-104">https://revise360.co.uk/experience.html?id=ns-104</a> |

## Big question, outcomes and success criteria

**Big question** — How can you recognise a phishing message, and what should you do about it?

### Prerequisites

The lesson before this one is ns - 103. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: ns - 101, ns - 102.

### Assessable outcomes

1. **ns-104-o1** — understand what phishing is
2. **ns-104-o2** — understand how to identify and protect against phishing

## Vocabulary, resources and preparation

### Key vocabulary — Phishing, Social engineering

### Resources

|                   |                                                |
|-------------------|------------------------------------------------|
| SLIDES            | presentations/NS_L04_PhishingInbox_Lesson.pptx |
| WORKSHEET         | worksheets/NS_L04_PhishingInbox_Worksheet.docx |
| ONLINE EXPERIENCE | experience.html?id=ns-104                      |

### Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

### How pupils get in

1. Go to `topics.html` on revise360.co.uk.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

**Worth knowing before you promise anything.** In this build APP\_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

## Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

| MIN   | PHASE                                     | TEACHER                                                                                      | PUPILS                                                                          | EVIDENCE TO CHECK                                                                      |
|-------|-------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 0-5   | <b>Retrieve</b>                           | Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites. | Answer the starter on the worksheet.                                            | The starter box on NS_L04_PhishingInbox_Worksheet.docx is filled in.                   |
| 5-8   | <b>Review the starter</b>                 | Take answers; correct the one most got wrong.                                                | Correct their own starter.                                                      | A correction written, not just a tick.                                                 |
| 8-18  | <b>Explain and model</b>                  | The new idea in two or three steps, with the example worked in front of the class.           | Listen and answer questions about the example.                                  | They can say why, not only what.                                                       |
| 18-21 | <b>Set up and settle</b>                  | Sign-in, launch, and what to write before anyone taps an answer.                             | Sign in, open the lesson, find What is phishing?.                               | Everyone is in and on station 1.                                                       |
| 21-43 | <b>Online: work the stations in order</b> | Pupils read each wall, write the key fact and the challenge, then answer at the badge.       | Read each wall, write the key fact and the challenge, then answer at the badge. | The worksheet has writing on it for each of the 6 stations before the badge is tapped. |
| 43-48 | <b>Check understanding</b>                | The hinge question, hands down.                                                              | Answer the hinge question, hands down.                                          | The split across the options.                                                          |
| 48-56 | <b>Apply</b>                              | Exam practice on the worksheet, in silence.                                                  | Exam practice, in silence.                                                      | worksheet, Exam practice a) to c)                                                      |
| 56-60 | <b>Exit and set up the next lesson</b>    | The exit question and the retrieval link to next lesson.                                     | Write the exit answer.                                                          | One sentence that answers the big question.                                            |

## The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

### 1. What is phishing?

- An online fraud technique: fake messages designed to trick people into giving away personal information.
- The message pretends to come from a trustworthy organisation, such as a bank or a school.
- It's a form of social engineering: it targets people, not software.

**Challenge on this wall:** Write the exam definition of phishing in your own words.

### 2. What attackers gain

- Usernames and passwords, which open email, banking and school accounts.
- Bank and card details, used for fraud or sold on.
- A way into an organisation's network, by stealing a staff login.

**Challenge on this wall:** What could an attacker do with a teacher's school password?

### 3. Spot the signs

- Check the sender's address, not just the display name.
- Look for urgency or threats: 'act now or your account will be closed'.
- Generic greetings, spelling mistakes and odd formatting are clues.
- Unexpected attachments or links asking you to log in.

**Challenge on this wall:** List four signs that an email might be phishing.

### 4. Check the link

- Hover over a link to see where it really goes.
- Attackers use lookalike addresses, such as barclays-secure-login.com.
- A padlock only means the connection is encrypted; it doesn't mean the site is honest.

**Challenge on this wall:** Which of the addresses on the wall is genuine, and how can you tell?

### 5. Targeted phishing

- Spear phishing targets one person, using details about them to look convincing.
- Whaling targets senior staff, such as a head teacher or finance director.
- Smishing uses text messages and vishing uses phone calls.

**Challenge on this wall:** Why is spear phishing more likely to succeed than a mass email?

### 6. What to do

- Don't click links or open attachments in suspicious messages.
- Check with the organisation using a number or address you already know.
- Report it: to the teacher, the IT team, or a national reporting service.
- If you did click, change your password straight away and tell someone.

**Challenge on this wall:** Write the three steps you would take if you thought you had been phished.

## Retrieval and the check for understanding

**Starter** — Think of a message you have received that looked suspicious. What made you doubt it? If you have never had one, describe what you would look for.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

### Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (ns-104-s4-c2), so the wording a pupil sees matches the wording you say.

**Does a padlock in the address bar mean a site is safe?**

**Correct:** No: it only means the connection is encrypted

### What a pupil might choose instead

- Yes, always
- Yes, if the site has a logo
- Only on phones

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

**The course's own response to a wrong answer here:** Criminals can get certificates for their fake sites too.

**One response, not four.** one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

## Guided to independent practice

| STAGE     | ACTIVITIES | WHAT IT IS                                                                                                                |
|-----------|------------|---------------------------------------------------------------------------------------------------------------------------|
| Recognise | 11         | The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.                   |
| Construct | 1          | Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer. |

### Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

### Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

## Delivering the online experience

### Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

### The stations, and what to record at each

| #     | STATION                         | ACTIVITIES | RECORD ON PAPER                                 |
|-------|---------------------------------|------------|-------------------------------------------------|
| 1     | What is phishing?               | 2          | Phishing is...                                  |
| 2     | What attackers gain             | 2          | A successful phish gives the attacker...        |
| 3     | Spot the signs                  | 1          | Signs of a phishing message include...          |
| 4     | Check the link                  | 2          | To check a link, you should...                  |
| 5     | Targeted phishing               | 2          | Targeted phishing includes...                   |
| 6     | What to do                      | 2          | If a message looks like phishing, you should... |
| final | Final challenge: phish or safe? | 1          | the answer to the station's challenge           |

### Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

### Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

## Assessment, evidence and unfinished work

| ID        | QUESTION                                                                                 | MARKS | WHERE                                                  |
|-----------|------------------------------------------------------------------------------------------|-------|--------------------------------------------------------|
| ns-104-x1 | Describe what phishing is and what the attacker is trying to obtain.                     | 3     | worksheet, Exam practice a); deck, Exam practice slide |
| ns-104-x2 | Give two features of a message that suggest it is phishing.                              | 2     | worksheet, Exam practice b); deck, Exam practice slide |
| ns-104-x3 | Explain two measures an organisation can take to reduce the risk of staff being phished. | 4     | worksheet, Exam practice c); deck, Exam practice slide |

### Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/ns-104.json`, which is not published with the website.

### Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What is phishing?** — Software that encrypts your files; Flooding a server with traffic
- **Why is phishing called social engineering?** — It only works on social media; It engineers new software
- **What is the main aim of a phishing attack?** — To slow down the victim's computer; To test the victim's antivirus
- **Why do attackers target staff logins at a company?** — Staff computers are faster; Staff emails are more fun to read
- **Which address is most likely to be genuine for a bank called Bank?** — `www.bank.co.uk.login.ru`; `www.bank-secure-login.com`

### Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

## Exit, next lesson and homework

### Exit check

One of these, written, not a show of hands:

- What phishing is
- What attackers gain
- Spotting and reporting phishing

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

### Retrieval link to next lesson

Next is ns-105, Social engineering HQ. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

### Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

## Shorter lessons, longer lessons and no internet

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### A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

### If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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